Universal Divisibility of Fixed-Length Consecutive-Integer Concatenations and OEIS A075018

Qichao Wang
Hebei University of Technology, Tianjin, China
Daoyu Dong
University of Electronic Science and Technology of China, Chengdu, China

October 17, 2026

Abstract

We prove that for every positive integer N there is a positive integer $k \geq 3$ such that the decimal concatenation $k \,\|\, (k-1) \,\|\, (k-2) \,\|\, (k-3)$ is divisible by N. Consequently, all terms of OEIS A075018 are positive. More generally, in any even integer base, for any prescribed number t of consecutive positive integers, any modulus N, and any specified permutation of the t blocks, one can find infinitely many same-digit-width concatenations divisible by N. The elementary constructive proof uses a base-supported/coprime factorization of N, a solvable linear congruence, and the Chinese remainder theorem. For the infinite family $N = d10^r$ with $d \mid 30$ and $r \geq 2$, we additionally determine the exact least value $\mathrm{A075018}(N) = N + 3$ and all admissible endpoints.

1 The OEIS problem

The sequence OEIS A075018 asks for the least integer $k \geq 3$ such that

$$N \mid k \,\|_{10}\, (k-1) \,\|_{10}\, (k-2) \,\|_{10}\, (k-3), \tag{1}$$

where N is the sequence index and the right-hand side is a *decimal concatenation*, not a product. The OEIS convention assigns zero if no such k exists [1]. We show that this zero case is impossible. The OEIS provides numerical values; a universal nonvanishing proof is not recorded in its entry at the time of preparation.

For positive integers $x_1, \ldots, x_t$ with exactly d base-b digits each, put $B = b^d$ and define

$$x_1 \,\|_b \cdots \|_b\, x_t = \sum_{i=1}^{t} x_i B^{t-i}. \tag{2}$$

Working within a single digit-length window is essential: it makes B constant, so the concatenation becomes an integer polynomial.

2 A general even-base theorem

Theorem 1 (Arbitrary modulus, length, and block order)**.** *Let $b \geq 2$ be an even integer and let $N, t \geq 1$ be independent positive integers. For every bijection $\pi : \{1, \ldots, t\} \longrightarrow \{0, \ldots, t-1\}$, there exist infinitely many pairs of positive integers (d, a) such that the consecutive integers $a, a+1, \ldots, a+t-1$ all have exactly d base-b digits and*

$$N \mid (a + \pi(1)) \,\|_b\, (a + \pi(2)) \,\|_b \cdots \|_b\, (a + \pi(t)). \tag{3}$$

In particular, the valid concatenated integers have unbounded length.

Proof. Factor $N = qm$, where q is the largest factor of N whose prime divisors divide b. Then

$$\gcd(q, m) = 1, \qquad \gcd(b, m) = 1, \qquad m \text{ is odd.} \tag{4}$$

Here $q = 1$ or $m = 1$ is allowed, with congruences modulo 1 understood as vacuous. Put $g = \gcd(t, m)$; because m is odd, g is odd and $g \mid t$. Hence

$$g \mid \frac{t(t-1)}{2}. \tag{5}$$

Indeed, $t(t-1)/2$ is an integer and twice this integer is divisible by g, while 2 is invertible modulo g. It follows from the standard linear congruence criterion that

$$ta + \frac{t(t-1)}{2} \equiv 0 \pmod{m} \tag{6}$$

has at least one solution a_0, and all its solutions constitute one residue class modulo $s = m/g$.

The last block in (3) is $a + \pi(t)$; impose

$$a + \pi(t) \equiv 0 \pmod{q}. \tag{7}$$

Since $\gcd(q, s) = 1$, the Chinese remainder theorem provides an integer a_* satisfying (6) and (7); all integers $a \equiv a_* \pmod{qs}$ satisfy both.

Choose an arbitrarily large digit width d with

$$q \mid b^d, \qquad b^d \equiv 1 \pmod{m}, \qquad (b-1)b^{d-1} \geq qs + t - 1. \tag{8}$$

These conditions can be met simultaneously: the first is automatic for all sufficiently large d; for $m > 1$ the second holds whenever d is a multiple of $\mathrm{ord}_m(b)$, and for $m = 1$ it is vacuous; the third holds eventually because $b \geq 2$. The interval of allowed first blocks

$$I_d = [b^{d-1},\, b^d - t] \cap \mathbb{Z}$$

then contains $(b-1)b^{d-1} - t + 1 \geq qs$ consecutive integers, so it contains an $a \equiv a_* \pmod{qs}$. Thus $a, a+1, \ldots, a+t-1$ are all positive and exactly d digits long.

Set $B = b^d$ and let C denote the right-hand concatenation of (3). Because $q \mid B$, the base-B expansion (2) gives

$$C \equiv a + \pi(t) \equiv 0 \pmod{q}.$$

Because $B \equiv 1 \pmod{m}$, the order of the blocks disappears:

$$C \equiv \sum_{i=1}^{t} (a + \pi(i)) = ta + \frac{t(t-1)}{2} \equiv 0 \pmod{m}.$$

As $\gcd(q, m) = 1$ and $N = qm$, we conclude $N \mid C$. The admissible d are arbitrarily large, producing distinct concatenations of unbounded length td. □

Remark 1. *The parity assumption on the base is used only to ensure that m is odd and hence (5) holds for every t. We do not claim the analogous unrestricted result for odd bases. The modulus N and the number of blocks t are* independent*; this differs from the special case $N = t$ of the even-base arithmetic-progression divisibility theorem [2].*

3 Explicit construction for A075018

Corollary 1. *For every $N \geq 1$ there are infinitely many integers $k \geq 3$ for which (1) holds. In particular,*

$$\boxed{\text{A075018}(N) > 0 \quad \textit{for all } N \geq 1.}$$

Proof. Apply Theorem 1 with $b = 10$, $t = 4$, and $(\pi(1), \pi(2), \pi(3), \pi(4)) = (3, 2, 1, 0)$. Then $k = a + 3$, so the concatenation is precisely $(a+3) \parallel (a+2) \parallel (a+1) \parallel a$. □

Here is a particularly simple algorithm not requiring a general linear-congruence solver. Write

$$N = qm, \qquad q = 2^{v_2(N)}5^{v_5(N)}, \qquad \gcd(m, 10) = 1.$$

As m is odd, 4 is invertible modulo m. The desired starting integer $a = k - 3$ may be chosen by

$$a \equiv 0 \pmod{q}, \qquad a \equiv -6 \cdot 4^{-1} \pmod{m}, \tag{9}$$

using the Chinese remainder theorem. Let a_* be its least nonnegative representative modulo N. Take any sufficiently large d satisfying

$$q \mid 10^d, \quad 10^d \equiv 1 \pmod{m}, \quad 9 \cdot 10^{d-1} \geq N + 3. \tag{10}$$

The least $a \geq 10^{d-1}$ congruent to a_* modulo N is

$$a = a_* + N \left\lceil \frac{10^{d-1} - a_*}{N} \right\rceil. \tag{11}$$

Then $10^{d-1} \leq a \leq 10^d - 4$ and

$$\begin{aligned} C &= (a+3) \,\|_{10}\, (a+2) \,\|_{10}\, (a+1) \,\|_{10}\, a, \\ C &\equiv a \equiv 0 \pmod{q}, \\ C &\equiv 4a + 6 \equiv 0 \pmod{m}, \end{aligned}$$

so $N \mid C$ and $k = a + 3$ is a witness. For example, $N = 7$ admits $a = 2$, giving $C = 5432 = 7 \cdot 776$ and A075018(7) = 5. The construction need not minimize k.

4 Exact least values for an infinite family

The general construction is not usually minimal. Nevertheless, for a natural infinite family the minimization problem is completely solvable, including a description of *all* admissible starting integers.

Theorem 2 (Exact minima and complete witnesses)**.** *Let $r \geq 2$ and let d be any positive divisor of* 30*, so $d \in \{1, 2, 3, 5, 6, 10, 15, 30\}$. Set $N = d10^r$ and $C(k) = k \,\|_{10}\, (k-1) \,\|_{10}\, (k-2) \,\|_{10}\, (k-3)$. Then for every integer $k \geq 4$,*

$$N \mid C(k) \quad \Longleftrightarrow \quad k \equiv 3 \pmod{N}.$$

Moreover, $C(3) = 3210$ is not divisible by N. Thus the complete set of admissible endpoints is $\{N + 3, 2N + 3, 3N + 3, \ldots\}$, and

$$\boxed{\text{A075018}(d10^r) = d10^r + 3 \quad (d \mid 30,\ r \geq 2).}$$

Proof. Suppose $k \geq 4$ and put $x = k - 3 \geq 1$. If $10^r \mid C(k)$, then the last r decimal digits of $C(k)$ must all vanish. Since its final concatenated block is the ordinary decimal representation of the positive integer x, this is possible only if x has at least r trailing zero digits. In particular, $x = t10^r$ for an integer $t \geq 1$.

Let ℓ denote the number of decimal digits of t, and let $U(k) = k \,\|_{10}\, (k-1) \,\|_{10}\, (k-2)$. Because the final block $x = t10^r$ has $\ell + r$ digits, we have the exact identity

$$\frac{C(k)}{10^r} = 10^\ell U(k) + t.$$

For the prime divisors 2 and 5 of d, the first summand is congruent to 0. Modulo 3, decimal concatenation is congruent to the sum of its constituent integers:

$$U(k) \equiv k + (k-1) + (k-2) = 3k - 3 \equiv 0 \pmod{3}.$$

Since every $d \mid 30$ is squarefree and has only prime divisors in $\{2, 3, 5\}$, it follows that

$$d \mid (10^\ell U(k) + t) \quad \Longleftrightarrow \quad d \mid t.$$

Consequently, $N = d10^r$ divides $C(k)$ exactly when $x = k - 3$ is a positive multiple of N. Finally, $C(3) = 3210 \equiv 10 \pmod{100}$, whereas every N in the statement is divisible by 100. So $k = 3$ is not admissible, and the first admissible endpoint is $N + 3$. □

For example, A075018(100) = 103, A075018(300) = 303, A075018(1500) = 1503, and A075018(3000) = 3003. The theorem does *not* assert a closed formula for A075018(N) at arbitrary indices N.

5 Verification and context

The accompanying Python program checks the 65 displayed terms of OEIS A075018 by direct search, validates constructive witnesses for many moduli, block counts, and even bases, and checks arbitrary permutations. These checks are independent of the universal proof.

The older OEIS A075000–A075002 work [2] deals with related arithmetic-progression concatenations, including divisibility by the *number of blocks*. The present theorem allows an *arbitrary modulus independent of the block count* and a prescribed ordering of the blocks. We make no claim that the stronger general theorem is novel before exhaustive prior-art review. Outside the exact family of Theorem 2, determining the least k remains a separate optimization problem.

References

[1] OEIS Foundation Inc., *A075018: Smallest k such that the concatenation $k, k-1, k-2, k-3$ is divisible by n*, https://oeis.org/A075018 (accessed October 17, 2026).

[2] Q. Wang, *Divisibility of Concatenated Arithmetic Progressions and Champernowne Prefixes* (2026), source and reference list at https://github.com/dongxuelian2/oeis-papers/tree/main/papers/A075000-A075002.

[3] F. Luca and B. Teguia Tabuguia, *Concatenations of terms of an arithmetic progression*, Journal of Difference Equations and Applications **32** (2026), no. 4, 437–456, https://doi.org/10.1080/10236198.2025.2603385.